Utilising Mobile Technology As a Tool to Supplementing Literacy Education For Underserved Children in Kenya

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Background

Uwezo (2016) indicated that a significant proportion of children in Africa do not have sufficient opportunities to learn English, which hinders their educational opportunities. In response to this issue, several non-profit organizations have provided English literacy programs as extra-curricular programs. However, results of those extra-curricular activities are unknown due to a lack of centralized evaluation efforts; additionally, learning outcomes and instructional strategies are not unified and inequitably distributed across geographical areas (Glewwe et al., 2009). Inspired by Kim et al. (2008), we attempted to utilize mobile technology in literacy education to see whether it can be a viable option to address critical educational challenges that are ultimately associated with a nation's workforce development and economy.

Objectives

To measure the practical functionality of mobile technology as a tool to assess students' literacy improvements, we ran an English training camp based on learners' literacy levels. The selected learners were involved in the boot camp for one-hour sessions for ten days after school. The learners were grouped according to their learning levels, as either Beginner, Letter, Word, Paragraph, or Story. The learners' progress was captured by pre- and post- assessments via a mobile-based literacy app. During the assessment, the students selected one of two paragraphs to read from and if successful, students moved on to the next section. Using Artificial Intelligence, the app was able to identify words mispronounced by learners during the assessment. The app used a custom Microsoft Azure speech-to-text model to convert reading audio of learners into text for evaluation. Moreover, the app follows the Uwezo literacy assessment protocol to evaluate and place learners in appropriate literacy learning levels.

Research Questions

1) Did literacy education using a mobile application measure improve learners' English literacy levels?
2) What learning environment factors might be related to learners' English literacy levels, including the number of family members, householder's education level, study assistance at home, and guardians' participation in school activities?

Method

The quantitative data employed in the study was derived from pre- and post- assessments in a literacy boot camp at Korando primary school and a survey of participants' learning environments. In both assessments, the Nyansapo AI was used whereby the child was presented with materials to read letters to a story. At the end of the boot camp, a short survey was used to collect demographic data, such as household size and education level.

Data Collection and Procedure

A total of 47 students (ages eight to 15; grade one to seven) were selected and participated in the camp through an initial assessment, and 44 students only completed the post-camp assessment and survey. Their literacy level was rated from one to six (new to advanced). Through the survey, students' learning environment factors were collected: age, number of family members, education level of a householder (primary = 1, secondary = 2, and university = 3), main guardian's participation in school activities (never = 1 to always = 4), study support from family or peers (never = 1 to always = 4), and attendance rate.

We used two types of statistical analysis: a paired t-test for pre- and post- assessments and bivariate correlation test. The results of the correlation test show the relationships between learning environment factors including the number of family members, householder's education level, study assistance at home (from mom, dad, sibling, and others), and guardian participation in school activities.

Results

The results of a paired sample t-test for pre- and post- assessments are shown in Table 1. The results indicated a statistically significant mean difference between pre- and post- assessment for learners' literacy levels at $p < 0.001$, showing a substantial increase in the learners' literacy level in the post- assessment. The mean value of learners' literacy increased substantially in post assessment with large effect sizes of 0.81 based on Cohen (1988).

Table 1 Means and standard deviations (SD) of pre- and post- assessment for students' literacy levels (N=44)

	Pre		Post		Mean Difference		
	Mean	*SD*	Mean	*SD*	(Pre-Post)	*t*	Cohen's *d*
Students' literacy levels	3.89	1.298	5.00	1.201	-1.114	-19.100**	.081

Note. ***$p \leq 0.01$. Cohen's *d* values were interpreted as: 0.20 = *small effect size*, 0.50 = *medium effect size*, and 0.80 = *large effect size* (Cohen, 1988).

Table 2 presents the results of correlations between learners' literacy levels in pre- and post- assessments and their learning environmental factors. As expected, pre- and post- assessments were highly and positively correlated. Also, *age* and performance in pre- and post- assessments were positively correlated. Interestingly, *the study assistant from mom* was negatively correlated with post- assessment. Additional positive correlations were found between: *age* and *number of family*, *age* and *study assistant from mom*, *guardian's participation* and *study assistant from mom* or *dad*, *study assistant from mom* and *study assistant from dad*, and *study assistant from siblings* and *study assistant from others*.

Results

Table 2 The result of Bivariate Correlation test for students' literacy level and their learning environment factors (N=44)

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11
1. Pre Test	3.89	1.30											
2. Post Test	5.00	1.20	**.955****										
3. Age	11.45	1.92	**.506****	**.574****									
4. Number of Family	5.89	1.40	.197	.276	**.417****								
5. Householder's education level	1.82	.58	.095	.100	.076	.031							
6. Guardian participation	2.89	.66	-.152	-.148	-.032	.036	-.056						
7. Study assistant from mom	2.84	1.03	-.274	**-.337***	-.197	-.157	-.204	**.488****					
8. Study assistant from dad	2.43	1.30	.181	.194	**.357***	.270	-.201	**.359***	**.468****				
9. Study assistant from sibling	2.98	1.05	.238	.278	.527	.183	.005	.393	.764	-.095			
10. Study assistant from others	2.02	1.39	.066	.070	-.021	.694	.033	.521	.175	.262	**.513****		
11. Attendance	90.40	14.47	-.109	-.156	-.123	**-.324***	.135	-.121	-.048	-.252	.026	.146	

Note. **$p \leq 0.01$ and *$p \leq 0.05$.

Conclusions

Considering 82% of the learners never had a chance to use a mobile device before, the resulting substantial increase in literacy levels in the post- assessment elucidated the effectiveness of literacy education measured by mobile technology. This result was aligned with Kim et al. (2008) who studied the possibility of implementing mobile technology in Latin America. As they emphasized, mobile technology can be a beneficial sustainable literacy instruction tool for developing countries. However, to implement mobile education, there are two linked realistic challenges that need to be solved. First, it is mandatory to establish an internet infrastructure in developing countries, along with continuous development of content applications. The current app was developed for assessment only. Second, if more contents for learning are developed, they can be utilized for after school programs according to students' levels in various subjects. Further, we recognized another benefit of the app-based literacy education: it will reduce exposure/contact to contagious diseases such as COVID-19, diarrhea, and tuberculosis which hinder students' learning compared to face-to-face traditional education.

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Background

Currently, a large number of children, in Africa, are experiencing a significant loss of educational opportunities because their daily conversational language is Swahili while they lack proficiency in the primary educational language, English. Due to this lack of proficiency, students are incapable of acquiring academic knowledge and skills. Uwezo (2016) indicated that a significant proportion of children in Africa do not have sufficient opportunities to learn English, which hinders their educational opportunities. In response to this issue, several non-profit organizations have provided English literacy programs as extra-curricular programs. However, results of those extra-curricular activities are unknown due to a lack of evaluation efforts; additionally, learning outcomes and instructional strategies are not unified and inequitably distributed across geographical areas (Glewwe et al., 2009). In order to promote children learning in Eastern Africa, it is imperative to reform current literacy education; however, there are unavoidable challenges involved in reforming literacy education, including a lack of adequately trained instructors and facilities (Kremer et al., 2013). Inspired by Kim et al. (2008), we attempted to utilise mobile technology in literacy education to see whether it can be a viable option to address critical educational challenges that are ultimately associated with a nation's workforce development and economy.

Objectives

To measure the practical functionality of mobile technology as a tool to assess students for the recruitment and their literacy improvements, we ran an English training camp based on learners' literacy levels. The selected learners were involved in the boot camp for one-hour sessions for ten days after school. The learners were grouped according to their learning levels, as either Beginner, Letter, Word, Paragraph or Story. The learners' progress was captured by pre and post assessments via a mobile-based literacy app. During the assessment, the students selected one of two paragraphs to read from and if successful, students moved on to the next section. Using Artificial Intelligence, the app was able to identify words mispronounced by learners during the assessment. The app used a custom Microsoft Azure speech-to-text model to convert reading audio of learners into text for evaluation. Moreover, the app follows the Uwezo literacy assessment protocol to evaluate and place learners in appropriate literacy learning levels. Everyday, the attendance of the children in both classes was also collected.

Research Questions

1) Was literacy education using a mobile application effective to improve learners’ English literacy levels?
2) What learning environment factors might be related to learners’ English literacy levels, including the number of family members, householder’s education level, study assistance at home, and guardian’t participation in school activities?

Method

The quantitative data employed in the study was derived from pre- and post- assessments in a literacy boot camp at Korando primary school and a survey of participants’ learning environments. In both assessments, the Nyansapo AI was used whereby the child was presented with materials to read letters to a story. At the end of the boot camp, a short survey was used to collect data about various elements external to the program such as factors affecting their households.

Data collection and Procedures

A total of 47 students (ages of eight to 15; grade one to seven) were selected and participated in the camp through an initial assessment, and 44 students only completed after-camp assessment and survey. Their literacy level was rated from one to six (new to advance). Through the survey, students’ learning environment factors were collected: age, number of family members, education level of a householder (primary = 1, secondary = 2, and university = 3) , guardian’s participation in school activities (never = 1 to always = 4), study support from family or peers (never = 1 to always = 4), and attendance rate.

We used two types of statistical analysis: a paired t-test for pre- and post- assessments and correlation. The results of correlation show relationship between learning environment factors including the number of family members, householder’s education level, study assistance at home (from mom, dad, sibling, and others), and guardian’t participation in school activities.

Results

The results of a paired sample t-test for pre and post assessments are shown in Table 1. The results indicated a statistically significant mean difference between pre and post assessment for learners’ literacy levels at $p < 0.001$, showing a substantial increase in the learners’ literacy level in the post assessment. The mean value of learners' literacy increased substantially in post assessment with large effect sizes of 0.81 based on Cohen (1988).

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Table 2 presents the results of correlations between learners' literacy levels in pre and post assessments and their learning environmental factors. As expected, *pre* and *post assessments* were highly and positively correlated. Also, *age* and performance in *pre* and *post assessments* were positively correlated. Interestingly, *the study assistant from mom* was negatively correlated with *post assessment*. Additional positive correlations were found between: *age* and *number of family*, *age* and *study assistant from mom*, *guardian's participation* and *study assistant from mom* or *dad*, *study assistant from mom* and *study assistant from dad*, and *study assistant from siblings* and *study assistant from others*.

Table 2 Correlation between students' literacy level and their learning environment factors (N=44)

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Note. **$p \leq 0.01$ and *$p \leq 0.05$.

Discussion

Considering 82% of the learners never had a chance to use a mobile device before, the resulting substantial increase in literacy levels in the post assessment elucidated the effectiveness of literacy education by mobile technology. This result was aligned with Kim et al. (2008) who studied the possibility of implementing mobile technology in Latin America. As they emphasized, mobile technology can be a beneficial sustainable literacy instruction tool for developing countries. However, to implement mobile education, there are two linked realistic challenges that need to be solved. First, it is mandatory to establish an internet infrastructure in developing countries, along with continuous development of applications and contents. The current app was developed for assessment only. Second, if more contents for learning are developed, they can be utilized for after school programs according to students' levels in various subjects. Further, we recognized another benefit of the app-based literacy education as if it will reduce exposure/contact to contagious diseases such as COVID-19, diarrhea, and tuberculosis which hinder students' learning compared to face to face traditional education.

References

Avery, R. J., Bryant, W. K., Mathios, A., Kang, H., & Bell, D. (2006). Electronic course evaluations: Does an online delivery system influence student evaluations? *The Journal of Economic Education*, *37*(1), 21–37. https://doi.org/10.3200/JECE.37.1.21-37

Uwezo,Kenya. (2016). Are our children learning. *Sixth Annual Learning Assessment Report*. https://twaweza.org//wp-content/uploads/2021/01/UwezoKenya2015ALAReport-FINAL-EN-web.pdf

Kim, P., Miranda, T., & Olaciregui, C. (2008). Pocket school: Exploring mobile technology as a sustainable literacy education option for underserved indigenous children in Latin America. *International Journal of Educational Development*, 28(4), 435-445.

Kremer M., Brannen C., Glennerster R. (2013) “The Challenge of Education and Learning in the Developing World.” Science, vol. 340 (6130), pp. 297–300.

Glewwe, P., Kremer, M., & Moulin, S. (2009). Many children left behind? Textbooks and test scores in Kenya. *American Economic Journal: Applied Economics*, *1*(1), 112-35.